THE KEEL-MORI MODULI SPACE IS LOCALLY OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4203-4215 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4217-4242, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 37 of queue_stacks-stacks-more-morphisms.json (pb=1171, stmt 4203-4215, proof 4217-4242).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00IS: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4242–4242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4242–4242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-etale-local-finite-inertia: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3798–3814. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-etale-separated-over-keel-mori: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4108–4116. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-well-nigh-affine: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3339–3346. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-well-nigh-affine-moduli-space: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3425–3433. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-keel-mori: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3854–3862. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $p : \mathcal{X} \to Y$ be a morphism of an algebraic stack to an algebraic space. Assume*

(1) *$\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is finite,*
(2) *Y is locally Noetherian, and*
(3) *p is locally of finite type.*

Let $f : \mathcal{X} \to M$ be the moduli space constructed in Theorem theorem-keel-mori. Then $M \to Y$ is locally of finite type.

Proof. Since f is a uniform categorical moduli space we obtain the morphism $M \to Y$. It suffices to check that $M \to Y$ is locally of finite type étale locally on M and Y. Since f is a uniform categorical moduli space, we may first replace Y by an affine scheme étale over Y. Next, we may choose I and $g_i : \mathcal{X}_i \to \mathcal{X}$ as in Lemma lemma-etale-local-finite-inertia. Then by Lemma lemma-etale-separated-over-keel-mori we reduce to the case $\mathcal{X} = \mathcal{X}_i$. In other words, we may assume $\mathcal{X}$ is well-nigh affine. In this case we have $Y = \mathrm{Spec}(A_0)$, we have $\mathcal{X} = [U/R]$ with $U = \mathrm{Spec}(A)$ and $M = \mathrm{Spec}(C)$ where $C \subset A$ is the set of R-invariant functions on U. See Lemmas lemma-well-nigh-affine and lemma-well-nigh-affine-moduli-space. Then A_0 is Noetherian and $A_0 \to A$ is of finite type. Moreover A is integral over C by Groupoids, Lemma 03BJ, hence finite over C (being of finite type over A_0). Thus we may finally apply Algebra, Lemma 00IS to conclude. □